

1. Introduction to Preliminaries

We introduce some basic structures and concepts from mathematics that are important to the study of Ordinary Differential Equation (ODE). While these concepts may not be necessarily part of the syllabus, they are briefly mentioned here and you should look into some advanced mathematics books from the library to get yourself familiarized with these concepts. Any advanced mathematics book on metric spaces or real analysis should contain more materials on these topics. I would urge you to write some proofs and try to understand the fundamental idea and motivation behind these concepts.

Definition (Field): Consider a set $\mathbb{K}$ with two binary operations $+\mathbb{K} : \mathbb{K} \times \mathbb{K} \rightarrow \mathbb{K}$ and $*\mathbb{K} : \mathbb{K} \times \mathbb{K} \rightarrow \mathbb{K}$ defined on $\mathbb{K}$. We will denote these binary operations using their infix notations i.e. $+\mathbb{K}(a, b)$ will be written as $a +\mathbb{K} b$ and $*\mathbb{K}(a, b)$ will be written as $a *\mathbb{K} b$. The operation $+\mathbb{K}$ is generally referred to as *addition* and $\alpha +\mathbb{K} \beta$ as the *sum of α and β* . The operation $*\mathbb{K}$ is generally referred to as *multiplication* and $\alpha *\mathbb{K} \beta$ as the *product of α and β* . The set $\mathbb{K}$ along with two such operations $+\mathbb{K}$ and $*\mathbb{K}$, denoted by $(\mathbb{K}, +\mathbb{K}, *\mathbb{K})$, will be considered a *field* if they satisfy the following axioms:

- For every pair of elements $\alpha, \beta \in \mathbb{K}$, there is a corresponding unique element $\alpha +\mathbb{K} \beta \in \mathbb{K}$ i.e. the operation $+\mathbb{K}$ is closed in $\mathbb{K}$.
 1. $\alpha +\mathbb{K} \beta = \beta +\mathbb{K} \alpha$ for all $\alpha, \beta \in \mathbb{K}$.
 2. $(\alpha +\mathbb{K} \beta) +\mathbb{K} \gamma = \alpha +\mathbb{K} (\beta +\mathbb{K} \gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{K}$.
 3. There exists an element $0_{\mathbb{K}} \in \mathbb{K}$ such that $0_{\mathbb{K}} +\mathbb{K} \alpha = \alpha$ for all $\alpha \in \mathbb{K}$. This element $0_{\mathbb{K}}$ is called the additive identity of the field.
 4. For every $\alpha \in \mathbb{K}$ there exists an element $\beta \in \mathbb{K}$ such that $\alpha +\mathbb{K} \beta = 0_{\mathbb{K}}$. The element β is called the additive inverse of the element α .
- For every pair of elements $\alpha, \beta \in \mathbb{K}$, there is a corresponding unique element $\alpha *\mathbb{K} \beta \in \mathbb{K}$ i.e. the operation $*\mathbb{K}$ is closed in $\mathbb{K}$.
 5. $\alpha *\mathbb{K} \beta = \beta *\mathbb{K} \alpha$ for all $\alpha, \beta \in \mathbb{K}$.
 6. $(\alpha *\mathbb{K} \beta) *\mathbb{K} \gamma = \alpha *\mathbb{K} (\beta *\mathbb{K} \gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{K}$.
 7. There exists an element $1_{\mathbb{K}} \in \mathbb{K}$ with $1_{\mathbb{K}} \neq 0_{\mathbb{K}}$ such that $1_{\mathbb{K}} *\mathbb{K} \alpha = \alpha$ for all $\alpha \in \mathbb{K}$. The element $1_{\mathbb{K}}$ is called the multiplicative identity of the field.
 8. For every $\alpha \in \mathbb{K}$ with $\alpha \neq 0_{\mathbb{K}}$ there exists an element $\beta \in \mathbb{K}$ such that $\alpha *\mathbb{K} \beta = 1_{\mathbb{K}}$. The element β is called the multiplicative inverse of the element α .
 9. The operation $*\mathbb{K}$ is distributive over $+\mathbb{K}$ $\alpha *\mathbb{K} (\beta +\mathbb{K} \gamma) = (\alpha *\mathbb{K} \beta) +\mathbb{K} (\alpha *\mathbb{K} \gamma)$ for all $\alpha, \beta, \gamma \in \mathbb{K}$.

If there is no ambiguity the field $(\mathbb{K}, +\mathbb{K}, *\mathbb{K})$, can be denoted just by the set $\mathbb{K}$.

Here are some examples of fields

1. The set of all rational numbers $\mathbb{Q}$ with the usual addition and multiplication operations forms a field, denoted as $(\mathbb{Q}, +, \times)$.
2. The set of all real numbers $\mathbb{R}$ with the usual addition and multiplication operations forms a field, denoted as $(\mathbb{R}, +, \times)$.
3. The set of all complex numbers $\mathbb{C}$ with the usual complex addition and complex multiplication operations forms a field, denoted as $(\mathbb{C}, +, \times)$.

In our study of ODEs we will keep our focus on the field of real numbers, $(\mathbb{R}, +, \times)$, and occasionally the field of complex numbers, $(\mathbb{C}, +, \times)$.

Definition (Vector Space): Consider a set $\mathbb{V}$ and a field $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$ with two binary operations $+_{\mathbb{V}} : \mathbb{V} \times \mathbb{V} \rightarrow \mathbb{V}$ and $*_{\mathbb{V}} : \mathbb{K} \times \mathbb{V} \rightarrow \mathbb{V}$ defined on $\mathbb{V}$. We will denote these binary operations using their infix notations i.e. $+_{\mathbb{V}}(a, b)$ will be written as $a +_{\mathbb{V}} b$ and $*_{\mathbb{V}}(a, b)$ will be written as $a *_{\mathbb{V}} b$. The set $\mathbb{V}$ along with any two operations $+_{\mathbb{V}}$ and $*_{\mathbb{V}}$, denoted by $(\mathbb{V}, +_{\mathbb{V}}, *_{\mathbb{V}})$, will be considered a *vector space over the field* $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$ if they satisfy the following axioms:

- For every pair of elements $a, b \in \mathbb{V}$, there is a corresponding unique element $a +_{\mathbb{V}} b \in \mathbb{V}$ i.e. the operation $+_{\mathbb{V}}$ is closed in $\mathbb{V}$.
- 1. $a +_{\mathbb{V}} b = b +_{\mathbb{V}} a$ for all $a, b \in \mathbb{V}$.
- 2. $(a +_{\mathbb{V}} b) +_{\mathbb{V}} c = a +_{\mathbb{V}} (b +_{\mathbb{V}} c)$ for all $a, b, c \in \mathbb{V}$.
- 3. There exists an element $0_{\mathbb{V}} \in \mathbb{V}$ such that $0_{\mathbb{V}} +_{\mathbb{V}} a = a$ for all $a \in \mathbb{V}$. The element $0_{\mathbb{V}}$ is called the *additive identity* of the vector space.
- 4. For every $a \in \mathbb{V}$ there exists an element $b \in \mathbb{V}$ such that $a +_{\mathbb{V}} b = 0_{\mathbb{V}}$. The element b is called the *additive inverse* of the element a . It is often written as $-a$.
- For every $\alpha \in \mathbb{K}$ and $a \in \mathbb{V}$, there is a corresponding unique element $\alpha *_{\mathbb{V}} a \in \mathbb{V}$.
- 5. $(\alpha *_{\mathbb{K}} \beta) *_{\mathbb{V}} a = \alpha *_{\mathbb{V}} (\beta *_{\mathbb{V}} a)$ for all $\alpha, \beta \in \mathbb{K}$ and $a \in \mathbb{V}$.
- 6. $1_{\mathbb{K}} *_{\mathbb{V}} a = a$ for all $a \in \mathbb{V}$ and where $1_{\mathbb{K}}$ is the multiplicative identity of the field $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$.
- 7. The operation $*_{\mathbb{V}}$ is distributive over $+_{\mathbb{V}}$ i.e., $\alpha *_{\mathbb{V}} (a +_{\mathbb{V}} b) = (\alpha *_{\mathbb{V}} a) +_{\mathbb{V}} (\alpha *_{\mathbb{V}} b)$ for all $\alpha \in \mathbb{K}$ and $a, b \in \mathbb{V}$.
- 8. The operation $*_{\mathbb{V}}$ is distributive over $+_{\mathbb{K}}$ i.e., $(\alpha +_{\mathbb{K}} \beta) *_{\mathbb{V}} a = (\alpha *_{\mathbb{V}} a) +_{\mathbb{V}} (\beta *_{\mathbb{V}} a)$ for all $\alpha, \beta \in \mathbb{K}$ and $a \in \mathbb{V}$.

If there is no ambiguity the vector space $(\mathbb{V}, +_{\mathbb{V}}, *_{\mathbb{V}})$ over the field $(\mathbb{K}, +_{\mathbb{K}}, *_{\mathbb{K}})$ can just be denoted by the set $\mathbb{V}$. The operation $+_{\mathbb{V}}$ is generally referred to as *vector addition* and $\alpha +_{\mathbb{V}} \beta$ as the *vector sum of α and β* . The operation $*_{\mathbb{V}}$ is generally referred to as *scalar multiplication* and $\alpha *_{\mathbb{V}} \beta$ as the *scalar product of α and β* .

Here are some examples of vector spaces

1. For any $n \in \mathbb{N}$ with $1 \leq n < \infty$, the set $\mathbb{R}^n$ or $\mathbb{R} \times \mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$ (n times), the set of ordered n -tuples of real numbers forms a vector space over the field $(\mathbb{R}, +, \times)$. The vector addition operation here is defined as the usual addition operation used component-wise as $(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$ and the scalar multiplication is defined as the usual multiplication operation defined component-wise as $\alpha \times (x_1, x_2, \dots, x_n) := (\alpha \times x_1, \alpha \times x_2, \dots, \alpha \times x_n)$ for all $\alpha \in \mathbb{R}$ and $(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$.
2. For any $n \in \mathbb{N}$ with $1 \leq n < \infty$, the set $\mathbb{C}^n$ or $\mathbb{C} \times \mathbb{C} \times \mathbb{C} \times \dots \times \mathbb{C}$ (n times), the set of ordered n -tuples of complex numbers forms a vector space over the field $(\mathbb{C}, +, \times)$. The vector addition operation here is defined as the usual complex addition operation used component-wise as $(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$ and the scalar multiplication is defined as the usual complex multiplication operation defined component-wise as $\alpha \times$

$(x_1, x_2, \dots, x_n) := (\alpha \times x_1, \alpha \times x_2, \dots, \alpha \times x_n)$ for all $\alpha \in \mathbb{C}$ and $(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in \mathbb{C}^n$.

- For any $n \in \mathbb{N}$ with $1 \leq n < \infty$, the set $\mathbb{C}^n$ or $\mathbb{C} \times \mathbb{C} \times \mathbb{C} \times \dots \times \mathbb{C}$ (n times), the set of ordered n -tuples of complex numbers forms a vector space over the field $(\mathbb{R}, +, \times)$. The vector addition operation here is defined as the usual complex addition operation used component-wise as $(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$ and the scalar multiplication is defined as the usual multiplication operation defined component-wise as $\alpha \times (x_1, x_2, \dots, x_n) := (\alpha \times x_1, \alpha \times x_2, \dots, \alpha \times x_n)$ for all $\alpha \in \mathbb{R}$ and $(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in \mathbb{C}^n$.
- For $a, b \in \mathbb{R}$ with $a < b$, consider the closed and bounded interval $[a, b] \subset \mathbb{R}$. The set of all continuous real-valued functions defined on the interval $[a, b]$ (any function $f : [a, b] \rightarrow \mathbb{R}$ which is continuous), denoted by $\mathcal{C}([a, b])$, forms a vector space over the field $(\mathbb{R}, +, \times)$ with the vector addition operation defined as the usual addition operation defined pointwise on $[a, b]$ as $(f + g)(x) := f(x) + g(x)$ and the scalar multiplication defined as the usual multiplication operation also defined pointwise as $(\alpha \times f)(x) := \alpha \times f(x)$ for all $x \in [a, b]$, $\alpha \in \mathbb{R}$, and $f, g \in \mathcal{C}([a, b])$.

There are many such vector spaces and I would urge you to look into any advanced mathematics text book on *Linear Algebra* for more examples and more information on vector spaces.

Definition (Normed vector Space): A vector space, $\mathbb{V}$ over the field $\mathbb{K}$, is a *normed vector space* $(\mathbb{V}, +_{\mathbb{V}}, *__{\mathbb{V}}; \|\cdot\|_{\mathbb{V}})$, if there exists a function $\|\cdot\|_{\mathbb{V}} : \mathbb{V} \rightarrow \mathbb{R}$ which satisfies the following conditions

- $\|x\|_{\mathbb{V}} \geq 0$ for all $x \in \mathbb{V}$ and $\|x\|_{\mathbb{V}} = 0$ if and only if $x = 0_{\mathbb{V}}$.
- $\|x +_{\mathbb{V}} y\|_{\mathbb{V}} \leq \|x\|_{\mathbb{V}} + \|y\|_{\mathbb{V}}$ for all $x, y \in \mathbb{V}$.
- $\|\alpha *__{\mathbb{V}} x\|_{\mathbb{V}} = \|\alpha\|_{\mathbb{K}} \|x\|_{\mathbb{V}}$ for all $\alpha \in \mathbb{K}$, $x \in \mathbb{V}$.

We call $\|\cdot\|_{\mathbb{V}}$ the *norm* of the vector space $\mathbb{V}$.

Definition (Metric Space): Let us consider a set $\mathcal{M}$ with $\mathcal{M} \neq \emptyset$ and a function $d : \mathcal{M} \times \mathcal{M} \rightarrow \mathbb{R}$, which satisfies the following conditions:

- $d(x, y) = d(y, x)$ for all $x, y \in \mathcal{M}$.
- $d(x, y) \geq 0$ for all $x, y \in \mathcal{M}$ and $d(x, y) = 0$ if and only if $x = y$.
- $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in \mathcal{M}$.

The function d is then called a *metric* and the set $(\mathcal{M}, d)$ is called a *metric space*. If there is no ambiguity we generally denote the metric space with just the set $\mathcal{M}$.

Exercise: Show that any normed vector space, say $(\mathbb{V}, +_{\mathbb{V}}, *__{\mathbb{V}}; \|\cdot\|_{\mathbb{V}})$, can also be considered a metric space with the metric defined as $d(x, y) := \|x - y\|_{\mathbb{V}}$.

Definition (Convergence): A sequence $(x_i)_{i=1}^{\infty}$ in a metric space $(\mathcal{M}, d)$, is said to converge to an element $x \in \mathcal{M}$ if $\lim_{i \rightarrow \infty} d(x_i, x) = 0$. We shall denote the point x as $\lim_{i \rightarrow \infty} x_i$.

Definition (Cauchy Sequence): A sequence $(x_n)_{n=1}^{\infty}$ in a metric space $(\mathcal{M}, d)$, is called a *Cauchy sequence* if for any $\varepsilon \in \mathbb{R}$ with $\varepsilon > 0$, there exists an $N_{\varepsilon} \in \mathbb{N}$ with $N_{\varepsilon} > 0$ such that $d(x_m, x_n) < \varepsilon$ for all $n, m \in \mathbb{N}$ with $n, m \geq N_{\varepsilon}$.

Definition (ε -neighborhood): Let us consider a metric space $(\mathcal{M}, d)$, an element $x \in \mathcal{M}$, and $\varepsilon \in \mathbb{R}$ with $\varepsilon > 0$. The set $\{z \in \mathcal{M} : d(x, z) < \varepsilon\}$ is defined as the ε -neighborhood of the point x .

Definition (open set): Let us consider a metric space $(\mathcal{M}, d)$ and a set $S \subseteq \mathcal{M}$. If for all $x \in S$ there exists an $\varepsilon \in \mathbb{R}$ with $\varepsilon > 0$ such that the ε -neighborhood of x is a subset of S , then S is called an *open set* and in this particular case an *open subset* of $\mathcal{M}$.

Definition (Complete Metric Space): A metric space $(\mathcal{M}, d)$ is called a *complete metric space* if all cauchy sequences in $\mathcal{M}$ converge to a point in $\mathcal{M}$.

Definition (Banach Space): A *Banach Space* is defined as a complete normed vector space.

1.1. What is an ODE?

Definition (Ordinary Differential Equation): Let us assume $I \subset \mathbb{R}$ is an open interval in $\mathbb{R}$, $t \in I$ is an independent scalar variable, $D \subset \mathbb{R}^{1+n(m+1)}$ is an open subset of $\mathbb{R}^{1+n(m+1)}$ with $x : I \rightarrow \mathbb{R}^n$ defined as an m -times differentiable function on I and $F : D \rightarrow \mathbb{R}^n$ defined as a continuous function on D . An Ordinary Differential Equation (ODE) is a relation of the form

$$F(t, x(t), \dot{x}(t), \dots, x^{(m)}(t)) = 0 \quad (1)$$

We consider $y : J \rightarrow \mathbb{R}^n$ an m -times continuously differentiable function defined on the open interval $J \subset I$ to be a *classical solution* of Equation 1 if $(t, y(t), \dot{y}(t), \dots, y^{(m)}(t)) \in D$ and Equation 1 holds for all $t \in J$.